ANUDEEP M

Full-Stack & Backend Developer

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SUMMARY

M.Tech Computer Science graduate (DSCE, Bengaluru) specializing in full-stack development with the TypeScript ecosystem: React and Tailwind frontends, Node.js/Express APIs, PostgreSQL, MongoDB, Prisma, Redis and Docker. Backend Development Intern with experience in RESTful APIs, request validation, error handling and MongoDB integration. Familiar with WebSockets for real-time features and gRPC, with hands-on RAG experience using LangChain, FAISS and local LLMs. Currently working on DurableFlow, a crash-proof workflow engine, and ReviewPilot, an AI pull-request reviewer.

SKILLS

Languages: JavaScript, TypeScript, SQL, Python, Java

Backend: Node.js, Express.js, gRPC, REST APIs, JWT, gRPC, WebSocket

Frontend: React, Next.js, Tailwind CSS

Databases and Messaging: MongoDB, PostgreSQL, Redis, Bull MQ, FAISS (vector store)

AI / ML: Mistral (Ollama), LangChain, Hugging Face

Core CS: DSA, OS, DBMS, OOP, Computer Networks, Distributed Systems

WORK EXPERIENCE

Backend Intern | Aquamenz Automation Pvt. Ltd., Bengaluru

Oct 2023 – Dec 2023

- Built **10+ RESTful API endpoints** using Node.js and Express.js to support application features and CRUD operations.
- Designed **5+ MongoDB collections/models** to structure application data and support efficient database operations.
- Optimized database queries to reduce average response time by **20%**, improving API performance under test conditions.
- Added request validation and centralized error handling, reducing recurring API errors by **15%** during testing.
- Automated **3+ background tasks** using BullMQ and Redis to handle asynchronous operations and reduce unnecessary blocking of request processing.

PROJECTS

ReviewPilot | AI-Powered Pull Request Reviewer

github.com/anudeep-mahabala/ReviewPilot | *Typescript · Node.js · Fastify · GitHub API · BullMQ · Redis · PostgreSQL · Prisma · LLM*

- **Architected** an AI-powered code review pipeline using TypeScript, GitHub API, and LLM structured outputs, evaluating **10 labelled bug-introducing pull requests** and measuring bug-detection recall of **80%** and **70%** finding precision across seeded defects.
- **Built** context-aware code analysis and hallucination filtering using Tree-sitter, diff parsing, and Zod validation, rejecting findings with invalid line references or unsupported code evidence and measuring false positives across **20 clean pull requests**.
- **Designed** an asynchronous review-processing system using Fastify, BullMQ, Redis, and PostgreSQL, testing duplicate-event handling and commit-based idempotency while benchmarking **p50 review latency, duplicate-review rate, and cost per pull request**.

DurableFlow — Crash-Proof Workflow Engine

github.com/anudeep-mahabala/DurableFlow | *Typescript · Node.js · Express.js · PostgreSQL · Prisma · Vitest*

- **Engineered** a durable workflow engine with PostgreSQL-backed step checkpointing and replay, recovering **82/100 simulated worker crashes** without re-executing previously checkpointed steps.
- **Coordinated 5** concurrent workers using lease-based coordination, heartbeat renewal, and fencing tokens on PostgreSQL transactions, preventing stale-worker updates during contention tests
- Built retry and idempotency mechanisms for external API operations, achieving zero duplicate business operations across 1,000 injected retry scenarios using idempotency-key validation and failure-injection tests.

Local RAG Fusion AI Assistant — Private Document Q&A Pipeline

github.com/anudeep-mahabala/local-rag | *Typescript · React · Node.js · Ollama · PostgreSQL*

- **Developed** a privacy-first RAG pipeline using TypeScript, Ollama, and local embeddings, processing documents into **800-character chunks with 100-character overlap**; verified local document ingestion, embedding generation, and LLM inference to keep document processing on-device.
- **Implemented** semantic retrieval and source-grounded answer generation using PostgreSQL/pgvector, cosine similarity search, and top-4 context retrieval; achieved **Recall 90%** across **20 questions** with known source pages and **66%** correctly supported answers in the evaluation set.
- **Created** streaming chat responses using React, TypeScript, and Server-Sent Events (SSE), with configurable retrieval settings and interchangeable LLM backends; measured **450ms** median time to first token, **90%** unsupported-question rejection rate, and **35ms** median retrieval latency across **20** test queries.

EDUCATION

M. Tech — Computer Science | CGPA: 8.9 / 10

Dayananda Sagar College of Engineering, Bengaluru · Dec 2024 – Sep 2026

B.E — Computer Science | CGPA: 7.5 / 10

BMS Institute of Technology, Bengaluru · Dec 2020 – May 2024

CERTIFICATIONS

- Google Cybersecurity Professional Certificate — Coursera
- React, Next.js & TypeScript Projects — Udemy (2025)
- Full-Stack Web Development Bootcamp — Udemy
- Data Structures in Python — Udemy
- Networking Fundamentals — Udemy

ACTIVITIES

- Organized and led 3+ technical events at college, coordinating logistics for 100+ participants per event
- Avid reader focused on systems design and engineering leadership literature